Alert-Budgeted Clinical Early Warning: Patient-Blocked Risk Calibration for Stateful Sepsis Alarm Policies

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Abstract

Clinical early warning must balance detection against a manageable alert load, yet discrimination metrics do not specify how often a deployed rule will alarm. We formulate early warning as a stateful alarm policy with a threshold, refractory window and escalation tier, and calibrate it against a patient-average daily alarm budget. The artificial-intelligence contribution is a reusable layer that converts a fixed risk score into an auditable policy: patient-level losses are combined with a finite-grid upper-confidence bound, a Learn-then-Test correction, a conformal risk control selector, a validation-ranked utility selector and a randomised mixture. The engineering application is a causal sepsis pipeline on 40,336 PhysioNet 2019 records from two hospitals with three scorers over three subject-disjoint seeds, an independent 2012 mortality stress test, official 2019 utility, decision-curve net benefit and a queue simulation. On site A, mean sensitivity rises from 0.185 at 0.25 alarms per patient-day to 0.774 at 2 per day. Applying the source threshold to site B lowers sensitivity at 1 per day from 0.585 to 0.315 despite a lower realised rate; conformal risk control raises B sensitivity to 0.482 at a higher rate, while the official utility reverses the ranking because it penalises false alarms; a utility-ranked variant avoids over-alarming at loose budgets, raising utility from 0.56 to 0.93 at 2 per day. The 2012 test shows that the budget travels while the operating point does not. These results support budget calibration as an engineering instrument, not a clinical claim.

Keywords: Clinical early warning, Sepsis onset, Alarm budgets, Stateful policies, Risk calibration, Clinical time series

1. Introduction

Deterioration events in hospitalised patients unfold over hours, and the physiological trajectory that precedes them is recorded long before the event itself is recognised. This temporal structure has motivated a substantial literature on data-driven early warning, ranging from static severity scores to sequential models that forecast deterioration trajectories from electronic health records [1] and compact, efficiency-aware networks that predict sepsis onset from structured time series [2]. Reported discrimination is high and improving, with in-hospital mortality and onset prediction now routinely exceeding the performance of classical scoring instruments [3]. The implicit promise of this line of work is that a sufficiently accurate model, once installed at the bedside, will shorten time to intervention.

Discriminative prediction and operational alerting address different questions. A score-to-alert conversion requires a decision rule, whereas threshold-independent ranking metrics do not specify how frequently that rule emits alarms. Reviews of intensive-care adverse-event prediction discuss incomplete reporting of alert burden, time-to-alert and deployment calibration [4]. Reviews of multivariate clinical time-series learning also motivate evaluation beyond architectural comparisons [5, 6]. Alert-fatigue studies reach the same conclusion from the workflow side: an alert that is statistically accurate but too frequent or too poorly targeted is dismissed, ignored or actively suppressed [7, 8], and intensive-care monitors can emit hundreds of alarms per patient-day, most of them clinically irrelevant [9]. These concerns support reporting the alarm policy and its realised load explicitly, rather than treating an area-under-the-curve comparison as a specification of bedside alerting behaviour.

Recent research improves the backbone that produces the risk score. For irregularly sampled and heterogeneously missing observations, approaches include reliability-aware temporal aggregation [10, 11], graph formulations over irregular observations [12], and masked-attention models for deterioration prediction [13]. Hybrid sequence-graph models report strong discrimination on the 2019 cohort [14], structure-preserving models attempt to keep each alert connected to its evidence [15], and operational early-warning indices are being developed outside the public benchmark [16]. Transfer-oriented variants align feature spaces across cohorts [17], while dynamic temporal graphs incorporate clinical calculators into the prediction pathway [18]. These methods motivate prospective backbone comparisons, but their adapters are not included in the current reference implementation. Our implemented contribution instead concerns the causal mapping from fixed scores to stateful alarms and the risk associated with selecting that mapping.

A separate literature attaches reliability estimates to clinical predictions. Selective prediction and calibration have been analysed for multimodal clinical classification [19], reliability estimation has been extended beyond output-space calibration for time-series classifiers [20], and conformal methods have been used for physiological regression uncertainty [21]. Conformal risk control extends coverage to a user-specified

expected loss [22], and recent work relaxes its monotonicity requirements and studies cross-validation variants [23, 24]. Time-series conformal methods also model temporal dependence directly [25, 26]. Our distinction is the unit and estimand of alarm-policy calibration: refractory state makes successive emitted alarms dependent, so those hours cannot simply be counted as independent patient samples. This does not imply that every conformal treatment of time series is invalid or that a stateful policy cannot be calibrated at the trajectory level. The reference construction uses independent patient blocks and fixed finite-grid upper-confidence bounds, a stronger sampling assumption than exchangeability alone.

This paper makes the alarm budget an explicit constraint and calibrates the policy that generates alerts. The implementation separates causal score estimation from full-trajectory policy replay. Its finite-sample bound concerns the population mean of patient-level rates, not a hard maximum for each patient or ward. We evaluate the procedure on the public 2019 A/B cohort with three subject-disjoint seeds, including a same-threshold external audit, a deliberately invalid timestep comparator and a target label-budget analysis. The contributions are as follows.

- We formulate early warning as a stateful alarm policy whose refractory and escalation parameters are control variables, so alarm volume becomes a design parameter rather than an emergent property of a threshold.
- We derive a patient-blocked finite-grid bound on the population alarm rate, compare it with a Learn-then-Test selector, a randomised mixture, a conformal risk control selector and a deliberately invalid timestep shortcut, and add a validation-ranked utility stage that chooses a calibration-feasible threshold without treating the budget as a target.
- We provide causal reference learners and cached-score drivers, and evaluate them under shared budgets on the 2019 A/B cohort, an independent 2012 mortality cohort, the official 2019 utility and a human-factors queue.
- We quantify target-label cost, benchmark LightGBM and partial NEWS/SIRS/qSOFA under the same budget, and report repeated-split coverage, PPV/NNA, ward load and ablations, so that conservatism and operational meaning are measured rather than asserted.

The remainder of the paper reviews related work (Section 2), develops the policy formulation and calibration procedure (Section 3), specifies the datasets and experimental protocol (Section 4), reports the results (Section 5), and discusses their scope and limitations (Section 6).

2. Related Work

2.1. Backbones for Irregularly Sampled Clinical Time Series

Recent work models irregular sampling explicitly, using continuous-time formulations [11], reliability-aware temporal aggregation [10], and graphs over observations and variables [12]. Hybrid sequence-graph designs report strong discrimination on the 2019 cohort [14], while structure-preserving sequence models attempt to keep each alert connected to its supporting evidence [15]. Complementary approaches model the observation process to synthesise irregular trajectories [27] or exploit informative missingness for ward deterioration prediction [13]. Domain-specific applications include venous thromboembolism detection after cancer treatment [28], while surveys synthesise the wider modelling landscape [29]. These methods are potential future score backbones, not models already integrated or reproduced in the current package. Any integration must retain prefix-only information and cannot reuse a full-trajectory imputation procedure without a separate causality audit.

2.2. Early Warning and Deterioration Prediction

Clinical early warning has moved from static severity scores toward dynamic, trajectory-aware prediction. Classical early-warning scores remain important baselines and have been externally validated in emergency and prehospital settings [30, 31], but they are typically fixed-weight instruments rather than calibrated policies. Recent multicentre work predicts the shape of sepsis deterioration rather than a single risk estimate, arguing that heterogeneous trajectories require patient-specific forecasting [1], while efficiency-oriented designs pursue compact models that remain explainable under the compute constraints of a ward workstation [2]. Others inject clinical reasoning into the prediction pathway: dynamic temporal graphs embed calculator-derived features so that the model inherits the semantics of established instruments [18], and language-model-guided simulation is used to generate interpretable physiological dynamics for early warning [32]. Transfer-oriented formulations address the practical reality that a model trained in one institution must be reused in another with different coding practice and case mix [17], and earlier work established that deep models over longitudinal inpatient records can outperform tabular baselines for outcome prediction [3, 33, 34]. Prospective implementation studies show that deployment changes the evidentiary question: a model that performs well offline may produce modest clinical effects and a substantial alert burden in practice [35]. Multisite and external validations of sepsis models likewise report performance degradation and calibration drift across hospitals [36, 37]. Across this literature the model is usually the unit of contribution; the alerting rule that converts its output into an operational signal is treated as an afterthought, and comparison between methods is almost always conducted at a fixed threshold rather than at a fixed alarm volume.

2.3. Calibration, Selective Prediction, and Conformal Guarantees

Calibration and selective prediction have been studied in multimodal clinical classification [19]. Reliability estimation for time-series classifiers also examines evidence beyond output remapping [20], while adaptation audits consider realistic

label delays [38]. Conformal risk control provides distribution-free control of an expected loss and has been extended to non-monotone losses and cross-validation variants [22–24]. Learn-then-Test makes the multiple-testing structure explicit: candidate configurations are null hypotheses, and a valid p-value turns rejection into a finite-sample risk certificate [39]; anytime-valid and quantile variants extend the same logic to optional stopping and tail risks [40, 41]. Time-series conformal methods handle temporal dependence explicitly [25, 26, 42], and prediction-set design has been studied for efficiency and human decision quality rather than coverage alone [43, 44]. Conformal methods have also been applied to physiological regression uncertainty [21], to performance estimation under distribution shift [45], and to patient-specific haemodynamic adaptation [46]. These contributions concern different units, targets and sampling assumptions; their guarantees cannot be declared invalid merely because an unrelated downstream alarm rule has memory. The present reference procedure instead specifies a patient-average alarm-rate estimand and uses independent-block concentration, without claiming the weaker assumptions or different risk statement of standard conformal constructions.

2.4. Evaluation Practice

Risk-of-bias instruments for prediction models address artificial-intelligence-specific issues in calibration, heterogeneity and applicability [47]. Reviews of intensive-care adverse-event and time-series prediction also discuss reporting and evaluation limitations [4–6]. The alert-fatigue literature adds a complementary measurement problem: dismissal, override and workflow burden are not captured by discrimination or calibration [7, 8, 48]. Industrial alarm management treats load, prioritisation and rationalisation as lifecycle metrics [49, 50]; clinical early warning has analogous goals but no shared patient-day rate budget. Data-sharing histories shape which cohorts are available for replication [51], and task heterogeneity complicates direct transfer of mortality models [52, 53]. These concerns motivate a protocol that reports requested budgets, realised rates, sensitivity and lead time alongside discrimination; shared budgets make the constraint explicit without implying identical realised volume.

3. Method

3.1. A Reusable Calibration Layer

The method is deliberately modular. The first component is a causal score model that maps the observed prefix of a trajectory to a risk score; the second is a stateful alarm policy that converts scores into a refractory and optionally escalating alarm sequence; the third is a patient-blocked loss that summarises the operational burden of that sequence; and the fourth is a finite-sample selector that certifies a threshold or returns an explicit no-alarm fallback. Only the score model is application-specific. The policy, loss and selector can be reused for any binary alarm task with a bounded per-patient event rate, including mortality, deterioration, arrhythmia and equipment alarms.

This separation makes the result an engineering framework rather than a particular sepsis pipeline. A hospital can replace the scorer, change the refractory window or add escalation without recomputing the certificate from scratch, provided the loss matrix and support bound are recomputed for the new policy. The deployment contract then has three separate objects: the budget is a property of the calibrated policy family, the certificate is a property of the calibration sample, and the realised ward load is a property of the arrival and handling model.

3.2. Problem Setting and Notation

Let patient i contribute observations $O_i = \{(t_{ij}, m_{ij}, v_{ij})\}_{j=1}^{n_i}$, with observation times t_{ij}, variable indices m_{ij}, and values v_{ij}. The number and spacing of observations may differ across patients and variables. Decisions are made on a prespecified regular grid with sampling interval Δ hours, using only observations available at each decision time. A causal score function maps this history to $s_{it} = f_\theta(O_{i,\le t}) \in [0, 1]$; any preprocessing, including imputation, must satisfy the same restriction.

An observed event has time $t_i^\star$, and a useful early alarm falls in $[t_i^\star - W, t_i^\star)$, where $W = 12$ hours is a design choice for the proposed evaluation. Records without an observed event are represented separately from records whose event time is outside observed follow-up. Clinical event time and the boundary of a shifted prediction label are not interchangeable. The data protocol in Section 4 therefore requires an explicit onset convention and does not evaluate lead time against an unobserved or left-censored clinical onset.

3.3. Stateful Alarm Policies

The policy $\pi_\lambda = (\lambda, \tau, \kappa)$ comprises a threshold, an integer refractory duration, and an integer escalation duration. Here $\tau \ge 0$ is the number of subsequent decision steps suppressed after an alarm; $\kappa = 0$ disables escalation, while $\kappa > 0$ permits an override after κ consecutive above-threshold steps following the last alarm. Let z_{it} be the number of refractory steps remaining at the start of step t, and let c_{it} include the current exceedance and the consecutive suppressed exceedances immediately preceding it. With initial state zero, the emission and update are

$$a_{it} = \mathbf{1}[s_{it} \text{ finite},\ s_{it} \ge \lambda,\ z_{it} = 0 \ \vee\ (\kappa > 0 \ \wedge\ c_{it} \ge \kappa)],$$
$$z_{i,t+1} = \begin{cases} \tau, & a_{it} = 1, \\ \max(0, z_{it} - 1), & a_{it} = 0. \end{cases} \tag{1}$$

The exceedance counter resets after an alarm or a non-exceedance, including a missing score. The refractory duration counts decision steps, not observations in the irregular raw record. Setting $\tau = 0$ gives the memoryless threshold policy, whereas $\lambda = +\infty$ gives an exact no-alarm policy. The latter is an explicit safe fallback and is distinct from the largest finite grid threshold, which can still emit alarms.

The scorer and structural parameters are selected using training and validation patients and are then fixed before risk calibration. Every candidate threshold is evaluated by replaying the entire causal policy, with state reset at each patient's admission. Within-patient alarm indicators are dependent, so

counting them as independent calibration units is not justified. Dependence within a patient does not itself prevent calibration over independently sampled patient trajectories.

3.4. Risk Estimand and the Timestep Shortcut

For n_i decision steps, define exposure $D_i = n_i\Delta/24$ days and alarm count $A_i(\lambda) = \sum_t a_{it}(\lambda)$. The target is the patient-average daily alarm rate

$$R(\lambda) = \mathbb{E}[A_i(\lambda)/D_i]\,. \tag{2}$$

This estimand weights each patient equally and is not the pooled ward rate $\mathbb{E}[A_i]/\mathbb{E}[D_i]$, which weights patients by exposure. Both may be clinically relevant, but their identification requires different loss definitions. The present certificate concerns only Equation (2); it does not guarantee a fixed maximum count for every patient or a fixed realised total for a ward.

The synthetic comparison includes a deliberately invalid effective-sample-size shortcut. It uses the same empirical patient-average losses as the blocked selector, but substitutes the number of calibration timesteps for the number of calibration patients in the confidence radius. This isolates the consequence of inflating the unit of information without changing the point estimator. It is not an implementation of a particular published timestep-conformal algorithm, nor does the comparison establish that every alternative time-series calibration method is invalid.

3.5. Patient-Blocked Uniform Risk Bounds

Define the patient loss as the observed daily alarm rate,

$$L_i(\lambda) = A_i(\lambda)/D_i \in [0, L_{\max}]. \tag{3}$$

The upper bound must hold over the deployment population's support, not merely over the observed calibration sample. If eligible records have at least $H_{\min}$ hours of exposure, let $n_{\min} = \lceil H_{\min}/\Delta \rceil$ and $g = \tau + 1$ without escalation, or $g = \min(\tau + 1, \kappa)$ when escalation is enabled. The minimum firing separation is g, so the exact discrete-support upper bound is

$$L_{\max} = \sup_{n\in\mathbb{N},\ n\geq n_{\min}} \frac{24}{n\Delta}\left(1 + \left\lfloor \frac{n-1}{g} \right\rfloor\right). \tag{4}$$

The rate decreases between firing-count increments, and its peaks at $n = 1 + qg$ decrease as q increases; hence checking $n_{\min}$ and the next such peak is sufficient. For hourly sampling, $\tau = 4$, no escalation and $H_{\min} = 24$, a 26-hour stay yields $L_{\max} = 144/26$, exceeding the five alarms per day possible over exactly 24 hours. Escalation can increase this bound and must not be omitted from its calculation. Losses outside the declared interval are rejected, not clipped, because clipping would certify a different estimand.

Condition on independently obtained training and validation data. The scorer, structural parameters, finite threshold grid Λ, and support bound are fixed under this conditioning. Calibration subjects are independent and identically distributed draws from the same population as future deployment subjects, although each subject's trajectory may have arbitrary internal dependence. The sample size n below is the number of calibration subjects, never the number of timesteps. Exchangeability alone is not sufficient for the independent-sample concentration argument used here.

Proposition 1 (Budget validity)**.** *Let* $\widehat{R}_n(\lambda) = n^{-1}\sum_{i=1}^n L_i(\lambda)$ *and* $\alpha \in (0,1)$*. Under the conditions above, select*

$$\widehat{\lambda} = \min\left\{\lambda \in \Lambda : \widehat{R}_n(\lambda) + L_{\max}\sqrt{\frac{\log(|\Lambda|/\alpha)}{2n}} \leq B\right\} \tag{5}$$

when the set is nonempty, and select the exact no-alarm policy $\widehat{\lambda} = +\infty$ *otherwise. For a finite nonnegative budget B, the selected policy satisfies*

$$\Pr\{R(\widehat{\lambda}) > B\} \leq \alpha. \tag{6}$$

The probability is over calibration samples; R is a population expectation conditional on the fixed scorer and policy family.

The proof is given in the supplement.

The empirical-Bernstein radius used in the reported experiments has a separate finite-grid guarantee. It is not an unproved tighter variant of Proposition 1.

Proposition 2 (Empirical-Bernstein validity)**.** *Under the fixed-scorer, finite-grid and independent-subject conditions of Proposition 1, let* $n \geq 2$ *and define*

$$\epsilon_{\mathrm{EB}}(\lambda) = \widehat{\sigma}_n(\lambda)\sqrt{\frac{2\log(2|\Lambda|/\alpha)}{n}} + \frac{7L_{\max}\log(2|\Lambda|/\alpha)}{3(n-1)}, \tag{7}$$

where $\widehat{\sigma}_n(\lambda)$ *is the sample standard deviation of* $\{L_i(\lambda)\}_{i=1}^n$ *with denominator* $n-1$*. Select*

$$\widehat{\lambda}_{\mathrm{EB}} = \min\left\{\lambda \in \Lambda : \widehat{R}_n(\lambda) + \epsilon_{\mathrm{EB}}(\lambda) \leq B\right\} \tag{8}$$

when the set is nonempty, and the exact no-alarm policy otherwise. Then

$$\Pr\{R(\widehat{\lambda}_{\mathrm{EB}}) > B\} \leq \alpha. \tag{9}$$

The proof is given in the supplement.

A sharper finite-sample alternative is conformal risk control [22]. It replaces the high-probability radius by an expected-risk correction and is particularly convenient for stateful policies because it requires only the bounded patient loss and exchangeability, not a variance estimate.

Proposition 3 (Expected-risk validity)**.** *Under the fixed-scorer, finite-grid and independent-subject conditions of Proposition 1, define*

$$\widehat{\lambda}_{\mathrm{CRC}} = \min\left\{\lambda \in \Lambda : \frac{n}{n+1}\widehat{R}_n(\lambda) + \frac{L_{\max}}{n+1} \leq B\right\} \tag{10}$$

when the set is nonempty, and the exact no-alarm policy otherwise. Then

$$\mathbb{E}\left[R(\widehat{\lambda}_{\mathrm{CRC}})\right] \leq B. \tag{11}$$

The proof is given in the supplement.

Proposition 4 (Randomised mixtures)**.** *Let $\lambda_1, \lambda_2 \in \Lambda$ satisfy $R(\lambda_1) \le B$ and $R(\lambda_2) \le B$, and let λ_p randomise between them with probability $p \in [0,1]$. Then $R(\lambda_p) = pR(\lambda_1) + (1-p)R(\lambda_2) \le B$. If the selector certifies the augmented family $\Lambda \cup \{\lambda_p\}$ with the same simultaneous bound, the randomised policy inherits the certificate.*

The proof is given in the supplement.

Proposition 3 is the expected-risk alternative used in Section 5.3; it is sharper than the blocked certificate but does not imply the high-probability statement of Proposition 2. Proposition 4 explains why a randomised selector can interpolate between grid points without leaving the certified family, provided the mixture itself is included in the family that is calibrated.

Proposition 1 is the conservative Hoeffding certificate, while Proposition 2 covers the empirical-Bernstein certificates reported in the experiments. The latter can be tighter when the per-subject alarm counts have low variance, but it is not uniformly tighter. Both are finite-grid upper-confidence risk procedures; subject blocking does not turn either into the standard expected-risk conformal construction. The manuscript's conformal terminology denotes this broader calibration motivation, not a claim that the proof requires only conformal exchangeability.

The same fixed-grid calculation can be read as a multiple-testing problem: each candidate threshold is a null hypothesis that its risk exceeds the budget, and a valid p-value turns rejection into a certificate. This is the Learn-then-Test construction [39]. A Holm step-down correction rejects at least as many hypotheses as the Bonferroni rule in Proposition 1 while retaining family-wise error control, so it is used as a less conservative comparator with the same guarantee. Recent anytime-valid variants relax the fixed-sample requirement [40], and quantile versions target a tail rather than the mean [41]; neither is required for the patient-average estimand studied here, but both clarify that the guarantee is a property of the testing procedure rather than of the score model.

The certificate always uses the raw empirical loss at the deployed threshold. Stateful alarm counts are not assumed to be monotone in the threshold, so isotonic regression is optional descriptive output and never substitutes for that loss in the upper bound. A finite test-cohort mean can exceed the budget even when its population expectation does not, so its exceedance frequency is an empirical diagnostic rather than the exact probability in Equation (6). Likewise, calibration transferred across distributions is not covered without the same-population assumption.

3.6. Operating Points and Lead-Time Evaluation

The reference selector deploys the smallest finite threshold whose raw risk upper bound meets the requested budget. This is a deterministic operating-point convention rather than a guarantee of optimal detection. Section 3.7 adds an optional validation-utility stage for settings in which the clinical cost of a false alarm makes the smallest feasible threshold a poor operating point. Neither a lower threshold nor a higher budget is assumed to improve every operational metric of a stateful policy.

3.7. Utility-Ranked Risk Calibration

An alarm budget constrains volume but does not encode the relative cost of a missed event and a false alarm. Let $U_i(\lambda)$ be patient i's contribution to the normalised 2019 utility at threshold λ, and let $\mathcal{V}$ be a validation partition disjoint from both calibration and test. For a calibration procedure that returns a feasible set $\mathcal{F}_B$ at budget B, define

$$\widehat{\lambda}_U = \arg\max_{\lambda \in \mathcal{F}_B} \left[\bar{U}_{\mathcal{V}}(\lambda) - z_{1-\delta} \frac{s_{\mathcal{V}}(\lambda)}{\sqrt{|\mathcal{V}|}} \right], \tag{12}$$

where $\bar{U}_{\mathcal{V}}$ and $s_{\mathcal{V}}$ are the validation mean and standard deviation of the patient-level contributions, and $z_{1-\delta}$ is the one-sided normal quantile at level $\delta = 0.10$. Ties are broken toward the lower validation alarm rate; if $\mathcal{F}_B$ is empty the selector returns the exact no-alarm fallback. The lower bound discourages a threshold whose apparent utility is driven by a few high-variance patients. It is not a finite-sample utility guarantee.

The risk interpretation depends on how $\mathcal{F}_B$ was constructed. With the blocked empirical-Bernstein certificate, every candidate in $\mathcal{F}_B$ is simultaneously certified on the calibration sample, so validation ranking cannot invalidate the high-probability risk statement. With CRC, the expected-risk guarantee is attached to the CRC selection rule itself; the utility-ranked threshold is a deployment variant, not a second certificate. We therefore report the CRC row and the utility-ranked CRC (U-CRC) row separately, and include the validation alarm rate so that a reader can see whether the utility stage consumed more of the budget than intended. This distinction is deliberate: a data-dependent choice among CRC candidates should not be presented as inheriting the CRC guarantee.

For an event observed during follow-up, define the detection window $\mathcal{W}_i = [\max(0, t_i^\star - W), t_i^\star)$. An event is detected if at least one alarm falls in that window, and its lead time is

$$\ell_i(\lambda) = t_i^\star - \min\{t \in \mathcal{W}_i : a_{it}(\lambda) = 1\}. \tag{13}$$

Lead time is undefined for undetected events and is represented as missing rather than zero. The reported median therefore describes detected events only and must always be accompanied by sensitivity and the number of eligible events. Events at admission, outside observed follow-up, or without an identifiable onset cannot supply an observed pre-onset lead time. A retrospective top-k selection over a complete trajectory is noncausal and may serve only as a labelled oracle comparator, not as a deployable early-warning policy.

Budget sweeps compare methods under the same requested upper-risk constraint, not necessarily at identical realised alarm rates. The plotted frontier uses realised alarm rate and conditional median lead time, retaining only nondominated pairs. At equal alarm rates only the largest lead time is retained; at larger rates a point remains only when its lead time strictly improves. Because conditional lead time can improve by missing difficult cases, sensitivity must be reported alongside the frontier before any clinical interpretation.

Algorithm 1 Patient-blocked upper-confidence calibration of an alarm policy

Require: independent calibration patients; fixed scorer, grid Λ, structural parameters, valid $L_{\max}$; budget B; level α

for $\lambda \in \Lambda$ **do**
    replay the causal policy on each patient's observed follow-up
    compute $L_i(\lambda)$ and reject any loss outside $[0, L_{\max}]$
    compute raw $\widehat{R}_n(\lambda)$ and Hoeffding or empirical-Bernstein radius
end for
$\mathcal{F} \leftarrow \{\lambda \in \Lambda : \widehat{R}_n(\lambda) + \epsilon(\lambda) \le B\}$
$\widehat{\lambda} \leftarrow \min \mathcal{F}$ if nonempty; otherwise $+\infty$
return threshold, raw certificate, and explicit fallback status

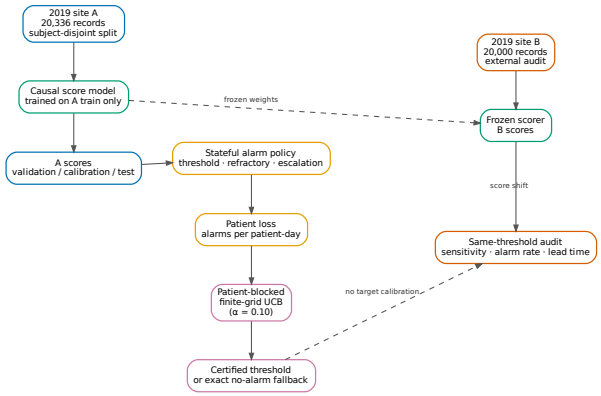


Figure 1: Evaluation design. A causal score model is fitted on site A training records and frozen before site B is touched. The stateful policy replays each patient trajectory to obtain alarms per patient-day; a patient-blocked finite-grid upper confidence bound selects a threshold, or returns the exact no-alarm fallback when no candidate is certified. The primary external audit applies the same threshold without target-site calibration; the target-calibration diagnostic in the Results is a separate oracle analysis.

4. Experimental Setup

4.1. Data Sources and Cohort Audit

Figure 1 summarizes the evaluation design. We use the public PhysioNet/CinC Challenge 2019 cohort [54], which contains 20,336 records from site A and 20,000 records from site B. The two sites are treated as distinct hospitals rather than interchangeable shards. All records were downloaded from the public release and checked against the published object sizes and MD5 entities before parsing. Site A contains 1,790 records with a positive sepsis label and 203 left-censored records whose first observed label is already positive; site B contains 1,142 positive and 223 left-censored records. After the six-hour label shift described below, 1,587 A records and 919 B records have an identifiable onset inside observed follow-up. Median observed stay is 39 hours at A and 38 hours at B.

The audit also exposes substantial measurement heterogeneity. In site A, $EtCO_2$ is absent in every record, whereas 7.6% of B records contain at least one $EtCO_2$ value; temperature is observed in only 33.8% of A records, and several laboratory variables are observed in fewer than 5%. This pattern motivates retaining a fixed variable set across sites and explicitly handling all-missing training columns, rather than fitting site-specific feature matrices. We also use the 2012 mortality cohort [55] as an independent outcome-shift stress test, described in Section 5.4; because it has no deterioration onset, it is not treated as a same-task sepsis validation.

4.2. Labels, Partitions and Eligibility

The supplied 2019 label becomes positive six hours before the challenge-defined clinical onset. We set the label lead to six hours to recover the candidate clinical time and a six-hour horizon to label the preceding window positive. Timesteps at or after onset, padded steps, and all steps of a left-censored record are excluded from discrimination and calibration targets. An event at or beyond the end of observed follow-up is not counted as a detectable lead-time case.

The official 2019 release stores one PSV file per subject, and we use "subject" and "patient" interchangeably for this cohort. We use the file stem as the subject identifier and verify that every identifier occurs in exactly one partition; the split audit is written to the run metadata. For each random seed we use a subject-disjoint 60/15/10/15 split into training, validation, calibration and test subjects. The scorer, preprocessing, structural policy parameters and threshold grid are fixed using training and validation subjects. The calibration partition selects the deployed threshold, and the test partition is touched only for evaluation. Site B is evaluated with the A-trained scorer and the A-calibrated threshold; no B labels are used for target calibration. Three seeds (0, 1, 2) are reported for the prefix logistic model, the masked GRU, the LightGBM baseline and the E2 budget sweep. The target-site recalibration diagnostic is reported for all three logistic seeds and for one GRU seed, and is labelled as an oracle analysis throughout.

A separate target-site diagnostic asks how much of the external loss is recoverable when target labels are available. We split the eligible B subjects into disjoint calibration and test halves of equal size using a site-specific random permutation (seed $1000 + s$ for source seed s), recalibrate the same policy family and threshold grid on B calibration subjects, and evaluate on B test subjects. This analysis uses target labels for threshold selection and is therefore an oracle-style target-calibrated reference estimate rather than a deployable label-free procedure; it is reported separately from the primary external audit and never used to tune the A model.

The target label-budget curve fixes the B test half and draws nested calibration subsets of 5%, 10%, 25% and 50% of all eligible B subjects from the remaining half. The same threshold selector is refitted on each subset and evaluated on the common test half, so the curve isolates calibration-set size. The source-threshold point is evaluated on the same test half and serves as the zero-label reference.

The support bound requires an independently declared minimum exposure. We use $H_{\min} = 24$ hours, which retains 15,837 of 20,336 A records and 14,824 of 20,000 B records. This is an

eligibility restriction, not a post-hoc filter chosen to improve results. Alarm-rate evaluation counts the full observed exposure, including time after onset; lead-time evaluation uses only the pre-onset window. With hourly decisions, refractory $\tau = 4$ and no escalation, the resulting support bound is $L_{\max} = 144/26 \approx 5.54$ alarms per patient-day.

4.3. Score Models

The primary scorer is a causal prefix-feature logistic regression. At each decision step it uses the last observed value of every variable, the cumulative observation fraction, and admission static features. All features are computed from current and past observations only. Median imputation is fitted on the training partition with empty features retained, so a variable that is entirely missing at site A remains a constant column at site B instead of silently changing the feature dimension. The classifier uses $C = 1$ and a maximum of 2,000 iterations. A unidirectional masked GRU with 64 hidden units is used as a secondary architecture check. It consumes observed values, observation masks, log-transformed elapsed time and static features; it is a reference implementation, not a GRU-D reproduction. The GRU is trained for 20 epochs with Adam at 10^{-3}, batch size 64, and a fixed random seed on an RTX 4090. A LightGBM gradient-boosted tree classifier is fitted to the same prefix features with 300 trees, learning rate 0.05, 31 leaves and median imputation; it is reported as a stronger tabular baseline, not as a tuned state-of-the-art model.

Partial NEWS, SIRS and qSOFA scores are computed from the variables available in the 2019 release, following the standard early-warning-score and instrument definitions [30, 56]. NEWS omits the consciousness and supplemental-oxygen components, SIRS omits the immature-neutrophil alternative, and qSOFA omits the Glasgow Coma Scale; these are research-only partial instruments, not certified clinical implementations. Charted observations are placed on the ICULOS clock, forward-filled, and scored per timestep. Steps before the first observed component receive zero points rather than being backfilled from a later observation, so the policy cannot use future measurements; the same no-points convention is applied to other unobserved steps. This is a missing-data convention, not an assertion that the patient was clinically normal. Their thresholds are calibrated on the same A calibration subjects at the same requested budgets as the learned scorers, so the comparison is a matched-workload audit rather than a claim that the instruments are equivalent.

4.4. Alarm Policy and Calibration

The policy family is the stateful rule in Equation (1). We fix refractory $\tau = 4$ decision steps, escalation $\kappa = 0$, and a 12-hour lead window $W = 12$. The candidate grid is 200 equally spaced thresholds from 0.01 to 0.99; the grid is fixed before calibration. The risk level is $\alpha = 0.10$, and the reported certificates use the empirical-Bernstein radius in Equation (7); Proposition 2 gives its finite-grid validity, while the Hoeffding radius in Proposition 1 is retained as the conservative variant. The selector deploys the smallest threshold whose raw empirical loss plus radius is at most the requested budget. If no threshold is certified, the exact no-alarm fallback is reported rather than silently substituting the largest finite threshold.

For each patient and threshold, the full stateful trajectory is replayed once to obtain alarms per patient-day. This loss matrix is reused across budgets and across the blocked and timestep procedures, so differences between procedures come only from the confidence radius and not from a different point estimator. The deliberately invalid timestep comparator substitutes calibration timesteps for calibration patients in the Hoeffding radius; it is a diagnostic, not a proposed method.

The finite-grid certificate is compared with five alternatives on the same loss matrix. A conformal risk control selector replaces the high-probability radius by the expected-risk correction in Proposition 3. A plug-in rule selects the smallest threshold whose empirical loss is at most the budget and has no finite-sample correction. A Learn-then-Test selector converts each candidate into a one-sided test and applies Holm or Bonferroni correction [39]. A randomised mixture draws one of two adjacent thresholds per patient; the mixture is certified over the augmented candidate family and can use the budget more fully than a single grid point. A validation-selected variant restricts attention to certified thresholds and ranks them by sensitivity on a disjoint validation partition. The certificate is always computed on calibration patients; validation is used only to break ties among already-certified candidates.

Repeated-split coverage is estimated with ten stratified partitions of the eligible cohort on each site, holding event and non-event patients separate. For each budget and procedure we report the fraction of splits whose realised mean rate exceeds the budget, the fallback rate, budget utilisation, sensitivity and the operational metrics. These split indicators estimate finite-cohort behaviour; they do not prove the population theorem, and their Monte Carlo resolution is reported alongside the point estimate.

Label-free target adaptation matches the requested rate on a disjoint warm-up half of the target site: the most aggressive threshold whose empirical rate is at most the budget is deployed, and sensitivity is evaluated on the held-out target half. This uses no target labels for threshold selection and is therefore a deployable heuristic, not a coverage guarantee. A ward simulation resamples observed per-hour alarm traces with Poisson arrivals, finite beds and a fixed hourly alarm-handling capacity; it translates per-patient rates into peak demand and the fraction of hours above capacity. One-at-a-time ablations vary the refractory window, escalation tier, lead window, minimum stay, risk level and threshold-grid resolution around the primary configuration.

4.5. Metrics and Uncertainty

Discrimination is summarized by AUROC and AUPRC over usable pre-onset timesteps. Because the positive-timestep prevalence is only 1.27% in the A test partition and 0.72% at B, AUPRC is interpreted against that prevalence rather than against 0.5. Calibration is summarized by 10-bin expected calibration error (ECE); ECE is descriptive and does not establish clinical calibration. Operational metrics are the realised patient-average alarm rate, event sensitivity in the 12-hour

lead window, median lead time among detected events, the fraction of test patients whose per-day alarm count exceeds the requested budget, positive predictive value (PPV), and the number needed to alert (NNA). The official 2019 utility and patient-level net benefit are computed on the same predictions; the alarm queue uses exponential service times with means of 5 and 15 minutes, following the response-time range reported by Bridi et al. [57].

Confidence intervals for score metrics use a patient-level bootstrap with 400 resamples, preserving within-patient dependence. Operational intervals use 2,000 patient-block resamples on seed 0 and resample whole target subjects, so within-subject alarm dependence is preserved; paired differences use the same resampled subjects and Holm correction across the reported comparisons. Two-sided bootstrap p-values are not a substitute for external validation. Operational curves report the mean and sample standard deviation over seeds; no interval is attached to the single-seed target-site GRU diagnostic.

4.6. Reproducibility and Compute

All raw records and score caches are kept outside the manuscript repository. The code exposes separate entry points for downloading and checksumming the 2019 release, auditing the official files, fitting and caching E1 scores, sweeping E2 budgets, bootstrap intervals and seed aggregation. The full test suite (172 tests) passes on the execution machine. The GRU uses CUDA but no formal GPU-hour or energy measurement was recorded; runtimes are therefore reported as implementation facts, not as a compute-efficiency claim.

5. Results

5.1. Cohort Audit and Score Discrimination

The public cohort is large enough to estimate patient-level alarm rates, but its measurement process is far from uniform. The supplementary cohort table shows that site A has a higher positive-record fraction than site B (8.8% versus 5.7%), while both sites contain a non-trivial number of left-censored records whose first label is already positive. The observed-variable table also explains why a naive cross-site model can fail: $EtCO_2$ is entirely absent at A but present in 7.6% of B records, and temperature is observed in only about one third of either cohort. A model that drops an all-missing training column would therefore change its input dimension at the external site; the implementation retains such a column as a constant during imputation and the regression test in the repository checks this case explicitly.

The full audit table is reported in the supplement.

Table 1 reports three-seed score discrimination. The prefix logistic model reaches an A-test AUROC of 0.736 ± 0.013 and AUPRC of 0.034 ± 0.006; the masked GRU reaches 0.743 ± 0.011 and 0.043 ± 0.005, and LightGBM reaches 0.763 ± 0.009 and 0.042 ± 0.004. These AUPRC values must be read against the 1.27% positive-timestep prevalence: both models rank positive steps several times better than chance, but the absolute positive predictive signal remains modest. On site B the logistic model falls to 0.676±0.005 AUROC and 0.017±0.000 AUPRC, while the GRU reaches 0.683 ± 0.004 and 0.018 ± 0.002, and LightGBM reaches 0.713 ± 0.003 and 0.025 ± 0.001. Ten-bin ECE is small in absolute terms (0.0011, 0.0017 and 0.0064 at A; 0.0014, 0.0018 and 0.0062 at B for logistic, GRU and LightGBM, respectively), but it is descriptive and should not be read as evidence of clinical calibration under shift. The drop is consistent with multisite sepsis-model evaluations that report degraded transport across hospitals [36, 37], and it is larger in AUPRC than in AUROC because prevalence also changes.

Table 1: Three-seed score discrimination (mean ± sample standard deviation). AUPRC is computed over usable pre-onset timesteps; positive-timestep prevalence is 1.27% at A and 0.72% at B. Ten-bin ECE is reported in the text for all three models and is descriptive.

	Site A test		Site B external	
Model	AUROC	AUPRC	AUROC	AUPRC
Prefix logistic	0.736 ± 0.013	0.034 ± 0.006	0.676 ± 0.005	0.0172 ± 0.0002
Masked GRU	0.743 ± 0.011	0.043 ± 0.005	0.683 ± 0.004	0.0183 ± 0.0016
LightGBM	0.763 ± 0.009	0.042 ± 0.004	0.713 ± 0.003	0.0250 ± 0.0010

The seed-0 patient bootstrap sharpens the comparison without turning a small numerical difference into a claim. At site A the logistic model has AUROC 0.741 with 95% interval $[0.710, 0.766]$ and AUPRC 0.041 with interval $[0.035, 0.051]$; the GRU has AUROC 0.744 $[0.716, 0.772]$ and AUPRC 0.044 $[0.038, 0.056]$. The paired GRU-minus-logistic difference is 0.004 for AUROC with interval $[-0.023, 0.032]$ and two-sided $p = 0.775$, and 0.003 for AUPRC with interval $[-0.006, 0.014]$ and $p = 0.42$. The GRU is therefore a secondary architecture check, not a demonstrated improvement. At site B the AUROC difference is essentially zero, while the AUPRC difference is 0.003 with interval $[0.0002, 0.0059]$ and $p = 0.05$; this isolated interval is reported as exploratory because it is not corrected across metrics or sites.

5.2. Budgeted Alarm Policies and Cross-Site Transport

Figure 2 and Table 2 report the operational frontier for the prefix logistic scorer. On site A, raising the requested budget from 0.25 to 2 alarms per patient-day increases mean sensitivity from 0.185 ± 0.061 to 0.774 ± 0.032, while the realised alarm rate rises from 0.108 ± 0.012 to 1.366 ± 0.026. The realised rate stays below the requested budget at every operating point, which is the expected consequence of a finite-sample upper bound rather than evidence that the policy is slack by design. The tightest budget has the largest seed-to-seed sensitivity variance, because a small change in the calibrated threshold moves several of the 168 eligible A events across the detection boundary. Median lead time among detected events stays between 8.7 and 10 hours, but this conditional median must be read with the corresponding sensitivity: it describes only the events that were detected.

The external audit exposes a qualitatively different operating point. At the 1-per-day budget the A-calibrated policy emits 0.276 ± 0.063 alarms per B patient-day, less than half the A rate, and detects 0.315 ± 0.062 of eligible B events compared with 0.585 ± 0.020 at A. At the tightest budget, B sensitivity

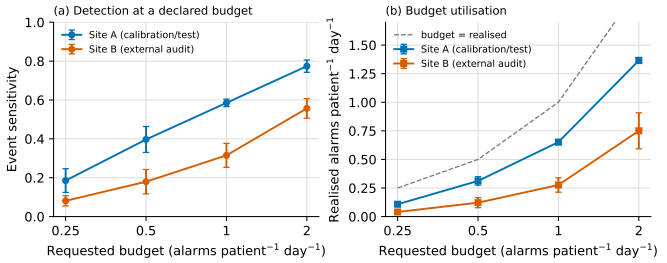


Figure 2: Three-seed E2 operating curves for the prefix logistic scorer. (a) Event sensitivity versus the requested patient-average alarm budget. (b) Realised alarm rate versus the requested budget; the dashed diagonal marks exact budget utilisation. Error bars are sample standard deviations over seeds 0–2. Site B uses the A-calibrated threshold without target labels.

Table 2: E2 budget sweep for the prefix logistic scorer, mean ± sample standard deviation over three seeds. Rates are alarms per patient-day; lead time is the median among detected events in hours; violation is the fraction of test patients whose per-day count exceeds the budget. The 4-per-day row is at the threshold grid floor; in the full sweep it coincides with 8 per day.

Site A (calibration/test)				
Budget	Rate	Sens.	Lead	Viol.
0.25	0.108 ± 0.012	0.185 ± 0.061	9.5 ± 0.5	0.074 ± 0.001
0.50	0.312 ± 0.037	0.397 ± 0.067	8.7 ± 0.6	0.151 ± 0.014
1.00	0.651 ± 0.022	0.585 ± 0.020	9.3 ± 0.6	0.210 ± 0.003
2.00	1.366 ± 0.026	0.774 ± 0.032	10.0 ± 0.0	0.309 ± 0.004
4.00	2.096 ± 0.026	0.894 ± 0.011	10.0 ± 0.0	0.271 ± 0.012

Site B (external audit)				
Budget	Rate	Sens.	Lead	Viol.
0.25	0.040 ± 0.014	0.081 ± 0.026	8.5 ± 0.5	0.030 ± 0.010
0.50	0.122 ± 0.044	0.180 ± 0.063	8.2 ± 0.3	0.065 ± 0.023
1.00	0.276 ± 0.063	0.315 ± 0.062	8.3 ± 0.6	0.094 ± 0.025
2.00	0.750 ± 0.157	0.557 ± 0.051	9.0 ± 0.0	0.163 ± 0.038
4.00	1.473 ± 0.250	0.724 ± 0.049	9.3 ± 0.6	0.131 ± 0.025

falls to 0.081 ± 0.026 while the realised B alarm rate is only 0.040 ± 0.014. The direction of the rate change is not a free lunch: the same threshold is simply less active on the shifted score distribution, and the lost activity is concentrated on true events. This is exactly the failure mode that a threshold-only transport claim would hide, and it is consistent with recent evidence that score distributions and calibration can shift across hospitals [45, 58]. A target-site adaptation step is therefore necessary before deployment, and the label-free rate-matching study below evaluates one such step without target labels.

To estimate how much of the transport loss is recoverable, we split target B into subject-disjoint calibration and test halves (7,412 patients each; about 354 calibration and 337 test events over three seeds) and recalibrate the same policy family on B labels. At 1 alarm per patient-day this oracle-style target calibration raises mean B sensitivity from 0.294 ± 0.082 to 0.497 ± 0.022 and the realised rate from 0.270 ± 0.065 to 0.616 ± 0.058; at 0.25 per day it raises sensitivity from 0.073 ± 0.026 to 0.218 ± 0.036. The recovery is real but partial and is purchased with a higher alarm load, so it identifies the value of target labels rather than a deployable procedure. The full table is in the supplement.

The timestep comparator is a diagnostic, not a proposed method, and the repeated-split study in Section 5.7 changes how it should be read. In the single split it selects a lower threshold than the blocked certificate at some budgets and raises sensitivity, but its realised mean rate remained below the requested budget in all ten repeated splits on both sites. It behaves like the plug-in rule because its effective sample size is large. The defensible criticism is therefore not that the shortcut always violates the mean-rate constraint; it is that the shortcut supplies no patient-level guarantee, and at 1 per day its patient-violation fraction is 0.296 versus 0.210 for the blocked certificate. A mean-rate bound estimated from correlated hours cannot certify the patient-average estimand, even when the realised mean happens to stay below budget on one cohort.

The GRU changes the level of the frontier but not its transport behaviour. At 1 alarm per patient-day its three-seed mean A sensitivity is 0.684 ± 0.006 and its B sensitivity is 0.375 ± 0.078, compared with 0.585 ± 0.020 and 0.315 ± 0.062 for the logistic model; the B alarm rate is also higher (0.328 ± 0.164 versus 0.276 ± 0.063). At 0.25 per day the GRU reaches 0.301 ± 0.082 at A and 0.115 ± 0.027 at B, again above the logistic frontier. The GRU is therefore a secondary architecture check rather than a demonstrated improvement: its seed-level spread is large, its paired site-B sensitivity interval includes zero after Holm adjustment, and its external alarm rate is higher. LightGBM is stronger on site B, but as the supplementary backbone comparison shows it still leaves the realised rate below the requested budget and does not remove the need for target calibration. A more expressive scorer can move the external frontier without repairing the site shift, and it can raise external alarm load while doing so. The calibration procedure, not the backbone alone, determines whether a stated alert budget is met.

Two limitations of the grid are visible in the table. First, the 4-per-day row already sits at the threshold grid floor and, in the full sweep, coincides with 8 per day; the grid stops at 0.01; above roughly 2 alarms per patient-day the budget constraint is no longer binding and the frontier is truncated by the candidate set. Second, every certified operating point has a realised rate below its requested budget, so the experiment demonstrates conservative risk control rather than tight budget utilisation. Neither observation undermines the certificate, but both matter for engineering deployment: a hospital that needs a precise alarm load may prefer a finer grid or a utility-based selection among certified thresholds, and a hospital that needs a higher sensitivity must accept the corresponding increase in realised alert volume. The repeated-split study in Section 5.7 shows that the empirical-Bernstein LTT and randomised-mixture selectors can raise utilisation without a mean-rate violation, so the conservatism of the blocked row is a property of that selector rather than of the budget constraint itself. The clinical interpretation of these rates also depends on local workflow and alarm-fatigue tolerance, which this retrospective study does not measure [7, 8].

Table 3: Blocked empirical-Bernstein versus conformal risk control on the primary A calibration split, with the A-calibrated threshold transported to B. "Cert." is the CRC certified risk on the calibration subjects; the request is the budget. CRC controls expected risk, whereas the blocked selector controls the probability of exceeding the budget.

	Internal A		External B		CRC	
Budget	Sens.	Rate	Sens.	Rate	Cert.	Status
0.25 blocked	0.250	0.117	0.106	0.052	–	feasible
0.25 CRC	0.363	0.239	0.200	0.115	0.240	feasible
1.00 blocked	0.607	0.631	0.382	0.348	–	feasible
1.00 CRC	0.667	0.905	0.482	0.549	0.974	feasible

5.3. A Sharper Selector with Conformal Risk Control

The blocked empirical-Bernstein certificate is valid but conservative. A conformal risk control (CRC) selector provides a sharper alternative for the same patient-blocked loss. For a candidate threshold λ, let $\widehat{R}_n(\lambda)$ be the mean calibration loss and let $L_{\max}$ be the support bound of Equation (4). CRC selects the smallest threshold satisfying

$$\frac{n}{n+1}\widehat{R}_n(\lambda) + \frac{L_{\max}}{n+1} \le B. \tag{14}$$

Under exchangeability of calibration and deployment subjects, the selected policy satisfies $\mathbb{E}[R(\widehat{\lambda})] \le B$ [22]. This is an expected-risk statement, not the high-probability statement of Proposition 2; the two certificates answer different questions and should not be compared as if one dominated the other.

Table 3 compares the blocked and CRC selectors on the primary A calibration split, with the same threshold grid and stateful replay. CRC more than doubles the internal sensitivity at the 0.25-per-day budget (0.250 to 0.363) and raises external B sensitivity from 0.106 to 0.200; at 1 per day it raises internal sensitivity from 0.607 to 0.667 and external sensitivity from 0.382 to 0.482. The gain comes from using more of the budget, not from a different score or a target-label oracle.

The repeated-split study in Section 5.7 confirms the pattern and exposes its cost. At 1 per day, CRC uses 93% of the budget on both sites and raises mean sensitivity from 0.613 to 0.706 at A and from 0.514 to 0.619 at B, while the blocked selector uses only 63%. At 0.25 per day, CRC uses 82% of the A budget and 93% of the B budget, compared with 75% and 65% for the blocked selector. The sharper selector is not uniformly aggressive: on the A 0.25 budget it still leaves 18% of the budget unused, and its mean rate exceeded the request in one of ten splits. That single exceedance is consistent with expected-risk control and is exactly why CRC should not be described as a PAC certificate. For deployment, the choice is therefore explicit: the blocked selector buys a high-probability budget guarantee at the cost of sensitivity, whereas CRC buys sensitivity at the cost of a weaker guarantee type and a higher patient-level violation fraction.

5.4. Independent-Cohort Stress Test

The 2019 A/B pair is a demanding but single transfer problem: both sites share the sepsis endpoint, the same challenge instrumentation and the same label construction. To ask whether the calibration layer travels beyond that pair, we apply the identical selector and policy family to the public PhysioNet/CinC Challenge 2012 cohort [55], whose endpoint is in-hospital mortality rather than sepsis onset. We use the three released hospital sets as A, B and C, retain records with at least 24 observed hours, and label the 12 hours preceding death as the detection window. The prefix logistic scorer is trained on set A and frozen; the A-calibrated threshold is then applied unchanged to sets B and C. This is an outcome-shift and instrumentation-shift stress test, not a sepsis replication, and we report it as such.

Table 4: Independent-cohort stress test on PhysioNet 2012. The scorer and threshold are calibrated on set A and transported unchanged to sets B and C; the endpoint is in-hospital mortality and the detection window is 12 hours. Rates are alarms per patient-day; "Util." is realised rate divided by the requested budget. At 0.25 per day no grid threshold is certified and the exact no-alarm fallback is returned on all three sets.

Cohort	Budget	Rate	Sens.	Util.
2012-A (calibration/test)	0.50	0.108	0.084	0.22
	1.00	0.416	0.313	0.42
	2.00	1.133	0.506	0.57
2012-B (external)	0.50	0.136	0.116	0.27
	1.00	0.461	0.324	0.46
	2.00	1.254	0.602	0.63
2012-C (external)	0.50	0.139	0.113	0.28
	1.00	0.465	0.303	0.47
	2.00	1.236	0.586	0.62

Three observations follow. First, the budget remains portable: no transported threshold exceeds its requested rate on either external hospital, and the realised rates remain in the same conservative band as the calibration site. Second, the transported operating point is not stable even when the budget is met. At 1 per day the realised rate rises from 0.416 on A to 0.461 and 0.465 on B and C, an 11–12% relative increase, while sensitivity changes only from 0.313 to 0.324 and 0.303. Third, the sharp sensitivity loss observed in the sepsis A→B transfer is therefore not a universal law of threshold transport. It is a property of that score, endpoint and cohort shift; with a different outcome and instrumentation, the same threshold becomes slightly more active and detects a similar fraction of deaths. The stable object is the calibrated budget, not a claim that transport always fails in one direction.

This distinction matters for how the paper's external result should be read. The sepsis audit shows that a source threshold can under-alarm on a target site; the 2012 stress test shows that the direction and magnitude of that error depend on the deployment distribution. A hospital cannot infer either sign from the source certificate, which is why the budget is the portable contract and target-site calibration or rate matching remains necessary. The test also has clear limits: mortality is not sepsis, the 2012 records use a different variable set and sampling grid, and the external sets are not independent of the challenge's original hospital grouping. We therefore treat it as a transport stress test that broadens the evidence base, not as a replacement for a prospective multi-hospital sepsis study.

Table 5: Official 2019 utility and decision-curve net benefit on the A calibration split and the transported B evaluation. "Utility" is challenge-normalised (inactive = 0, oracle = 1); $NB_{0.10}$ is patient-level net benefit at $p_t = 0.10$. At 0.25 per day CRC improves utility over the blocked certificate; at 1 per day the blocked selector is better because CRC buys sensitivity with false alarms.

		Internal A		External B	
Budget	Policy	Utility	$NB_{0.10}$	Utility	$NB_{0.10}$
0.25	Blocked	0.505	0.0206	0.142	0.0053
0.25	CRC	0.687	0.0244	0.247	0.0074
0.25	LTT	0.550	0.0229	0.185	0.0060
1.00	Blocked	0.932	0.0236	0.278	0.0053
1.00	CRC	0.852	0.0203	0.154	0.0007
1.00	LTT	0.932	0.0236	0.278	0.0053

5.5. *Official Utility and Decision-Curve Analysis*

Sensitivity under a rate budget does not encode the clinical cost of a false alarm. We therefore evaluate the same deployed policies with the official PhysioNet 2019 utility, which rewards a true alarm near the six-hour optimum, penalises a missed event and subtracts a fixed cost for each false alarm [54]. The score is normalised so that inactivity is zero and the oracle is one. For a patient-level decision curve we also report net benefit at threshold probabilities $p_t = 0.05, 0.10, 0.20$, where net benefit is $\text{sens} \cdot \pi - (1 - \text{spec})(1 - \pi)p_t/(1 - p_t)$ and π is event prevalence.

The official utility is not monotone in the requested budget. At 0.25 per day, CRC raises normalised utility from 0.505 to 0.687 at A and from 0.142 to 0.247 at B, because the blocked certificate leaves so much budget unused that the extra true alarms dominate the false-alarm cost. At 1 per day the ordering reverses: the blocked selector reaches 0.932 at A and 0.278 at B, whereas CRC falls to 0.852 and 0.154. CRC is optimising sensitivity subject to a rate constraint, not utility, and at the looser budget its additional alarms are mostly false. The decision curve tells the same story at $p_t = 0.10$: CRC is better at 0.25 (0.0244 versus 0.0206 at A; 0.0074 versus 0.0053 at B) and worse at 1.0 (0.0203 versus 0.0236 at A; 0.0007 versus 0.0053 at B). At $p_t = 0.20$ every policy has negative net benefit at B, so the clinical case for alarm-based screening weakens when the decision threshold is high.

This tension is an engineering result, not a defect of either selector. A hospital that wants the greatest chance of catching an event under a hard alarm budget should use CRC at tight budgets; a hospital that wants to maximise the official utility at a looser budget needs a utility-aware operating point rather than the smallest feasible threshold. The certificate constrains the alarm rate, but it does not choose the point on the utility curve. Reporting both quantities prevents a sensitivity gain from being presented as a clinical improvement when it is purchased with false alarms that the utility penalises; Section 5.6 turns this observation into a selector.

5.6. *Utility-Ranked Selection*

The utility reversal in Table 5 is not a reason to abandon calibration; it is a reason to separate the rate constraint from the operating-point objective. We therefore rank the CRC-feasible thresholds by validation utility using Equation (12). Table 6 reports the resulting U-CRC selector alongside the raw CRC and blocked rules. At 0.25 and 0.5 per day, U-CRC selects the same threshold as CRC and changes utility only in the third decimal. At 1 and 2 per day, it declines to spend the full budget: the 2-per-day request selects the same 0.0248 threshold as the 1-per-day request, with an internal rate of 0.631 rather than 1.355 and an external rate of 0.348 rather than 0.932. Internal utility rises from 0.562 to 0.932 and external utility from −0.225 to 0.278.

Table 6: Official 2019 utility for the blocked, raw CRC and utility-ranked CRC selectors on the primary seed. Internal A is the held-out test partition; external B is the transported evaluation. U-CRC ranks the CRC-feasible thresholds by a one-sided validation utility bound (Equation (12)); it is a deployment variant of CRC, not a second risk certificate.

	Internal A utility			External B utility		
Budget	Blocked	CRC	U-CRC	Blocked	CRC	U-CRC
0.25	0.505	0.687	0.691	0.142	0.247	0.244
0.50	0.778	0.904	0.910	0.266	0.273	0.270
1.00	0.932	0.852	0.932	0.278	0.154	0.278
2.00	0.562	0.562	0.932	−0.225	−0.225	0.278

The loose-budget row is the main engineering result. A budget is an upper bound, not a target: when CRC treats 2 alarms per patient-day as a mandate, the additional alarms are overwhelmingly false and the official utility falls below the 0.25-per-day operating point. U-CRC detects this with validation utility and stops raising the alarm load even though the calibration certificate would permit it. The same pattern appears at 1 per day, where U-CRC recovers most of the blocked selector's utility while retaining the sharper CRC candidate set at tighter budgets. The result is not a universal dominance claim: on the primary seed U-CRC and blocked are identical at 1 per day, and across seeds the 1-per-day difference ranges from a tie to a 0.028 loss. The stable part is the tight-budget gain and the refusal to over-alarm at 2 per day.

Two caveats follow from the construction. First, the validation lower bound reduces variance-driven selection but does not provide a finite-sample utility guarantee; utility is an evaluation objective, not a certificate. Second, at 0.25 per day the primary seed's validation rate is 0.273 while the test rate is 0.239, so the validation partition is a diagnostic rather than a second budget guarantee. The raw CRC row has the same validation rate at that threshold, so the discrepancy is sampling variation in the validation split, not a side effect of utility ranking. A deployment that requires a high-probability budget statement should retain the blocked selector; a deployment that treats the budget as a workload ceiling and cares about false-alarm cost can use U-CRC with both rates reported.

5.7. *Repeated-Split Coverage and Budget Utilisation*

A single calibration/test split cannot estimate how often the certificate is violated. We therefore repeat the patient-level split ten times on each site, calibrate on half of the eligible subjects and evaluate on the other half, and record the realised mean

Table 7: Repeated-split coverage at the 1-per-patient-day budget, ten stratified splits per site. "Rate" is the mean realised alarms per patient-day, "Util." is rate divided by 1.0, and "Patient viol." is the mean fraction of test patients above the per-patient budget. Blocked, LTT and randomised selectors had no split above the mean budget; CRC crossed it in one of ten A splits at 0.25 per day, consistent with its expected-risk rather than PAC guarantee. Blocked is the empirical-Bernstein selector; LTT is Learn-then-Test with Holm correction; Random. is the randomised adjacent-threshold mixture.

	Site A			Site B		
Procedure	Rate	Util.	Patient viol.	Rate	Util.	Patient viol.
Blocked	0.634	0.63	0.210	0.630	0.63	0.215
CRC	0.931	0.93	0.296	0.935	0.93	0.308
Timestep	0.931	0.93	0.296	0.935	0.93	0.308
Plugin	0.931	0.93	0.296	0.935	0.93	0.308
LTT	0.872	0.87	0.279	0.935	0.93	0.308
Random.	0.878	0.88	0.284	0.896	0.90	0.302

rate, the fraction of splits above the requested budget, the no-alarm fallback rate and the patient-level violation fraction. The supplement plots budget utilisation; Table 7 reports the 1-per-day row.

Three findings follow. First, the blocked, LTT and randomised certificates exceeded the mean budget in none of the ten splits on either site; the plug-in rule and CRC each crossed it in one of ten A splits at the 0.25-per-day budget, where the plug-in empirical rate was 0.221 against a request of 0.25 on average. That single CRC exceedance is the expected-risk signature discussed in Section 5.3, not a failure of the blocked certificate. Second, the timestep shortcut also stayed below the mean budget in all splits. The data therefore do not support the stronger claim that the shortcut always violates the mean-rate constraint; its deficiency is that it has no patient-level guarantee, and at 1 per day its patient-violation fraction is 0.296 versus 0.210 for the blocked certificate. It behaves like the plug-in rule because its effective sample size is large, so it selects the most aggressive empirical threshold. Third, budget utilisation is the main engineering cost of the certificate. At 1 per day the blocked selector uses 63% of the budget on both sites, whereas CRC uses 93% and raises sensitivity from 0.613 to 0.706 at A and from 0.514 to 0.619 at B. Across the four budgets the blocked selector uses 63–75% at A and 63–88% at B; CRC uses 82–93% at A and 93% at B. The empirical-Bernstein LTT and the randomised mixture occupy an intermediate band, at the price of a higher patient-level violation fraction. The plug-in rule reaches 88–93% on site A at the three tighter budgets and 93% at 0.25 on site B, but only 70% at 0.5 and 88% at 2 on site B; it has no finite-sample guarantee and exceeds the budget in one of the ten A splits at 0.25. This is the trade that the original single-split table could not show: the certificate is conservative, and the conservatism can be reduced by a sharper selector or by randomisation, not by pretending that the point estimate is a certificate.

5.8. Operational Alarm Metrics

Sensitivity and mean rate do not specify how many alarms a clinician sees for each detected event. We therefore report positive predictive value (PPV), the number needed to alert (NNA), and alarms per detected event, all computed from the same replay as the policy outcomes. On the primary A split at 1 per day, the blocked logistic policy emits 0.631 alarms per patient-day with PPV 0.060 and NNA 16.7; the plug-in policy emits 0.905 alarms per day with PPV 0.050 and NNA 20.0. On the external B evaluation the corresponding blocked values are PPV 0.037 and NNA 27.2, and the plug-in values are PPV 0.032 and NNA 31.4. At the tighter 0.25-per-day budget the blocked policy has PPV 0.099 and NNA 10.1 at A, because the same number of true alarms is divided by far fewer false alarms. The absolute PPV is low because the event prevalence is low and a patient can generate many hourly alarms; the useful engineering statement is the monotone trade: a tighter budget raises PPV and lowers NNA, while a looser budget buys sensitivity with more false alarms per detected event. These metrics also explain why the same sensitivity can correspond to very different clinical workloads across sites.

5.9. Target Adaptation in Brief

A target site that can observe unlabelled traces can match its requested rate without outcome labels. On a disjoint warm-up half of B, the most aggressive threshold whose empirical rate is at most the budget is deployed on the held-out half. At 1 per day this raises sensitivity from 0.414 to 0.620 and the realised rate from 0.356 to 0.950; at 0.5 it reaches 0.414 at 0.356, coinciding with the target-calibrated oracle on this grid. The rule is transductive and controls the rate rather than the risk, so it is a deployment heuristic rather than a replacement for a labelled certificate. The supplement reports the full adaptation and ablation tables.

5.10. Ward-Level Alarm Load

Per-patient rates are difficult to translate into staffing. We replay the calibrated traces on a synthetic 20-bed ward with Poisson arrivals, finite beds and a fixed hourly alarm-handling capacity. At five admissions per day and a capacity of one alarm per hour, the blocked policy exceeds capacity in 0.2% of hours at 0.25 alarms per patient-day, 3.8% at 1 per day and 12.7% at 2 per day. At a capacity of two alarms per hour the corresponding fractions are 0.0%, 0.2% and 1.6% (see the supplementary operational figure). The peak hourly demand rises from two to three alarms, and the 95th percentile from one to two. At ten admissions per day the ward reaches 84% bed occupancy and refuses 13.6% of admissions at the 2-per-day operating point, so the load estimate is conditional on the arrival and capacity assumptions rather than a property of the policy alone. The simulation is a translation device, not a validation: it shows that a budget difference of a factor of two is operationally visible at the nursing-station level, and that a ward with one alarm per hour of spare capacity is stressed even at 1 alarm per patient-day.

A continuous-time queue provides a human-factors check on the hourly capacity abstraction. Using the 5-minute median response reported by Bridi et al. and a 15-minute mean that includes travel and confirmation [57], one nurse handles the blocked 1-per-day policy with fewer than 2% of alarms waiting more than 10 minutes. At 2 per day the same nurse leaves

9.4% of alarms waiting more than 10 minutes, with a 95th-percentile wait of 26.8 minutes; a 5-minute mean reduces the over-10-minute fraction to 0.5%, and a second nurse removes essentially all waiting. The full queue table is in the supplement. These delays are lower than the 60% unanswered rate in the observational study because the queue assumes perfect work-conserving service; it is a lower bound on human delay, not a validation of the ward model. Even so, the difference between a 1- and 2-per-day budget is visible in response latency at average nurse utilisation below 20%, because alarm bursts matter more than the daily mean.

6. Discussion

The central empirical finding is that a patient-blocked alarm budget is portable as a *constraint* but not as a *threshold*. Within site A, the calibrated policy moves smoothly from 0.19 sensitivity at 0.25 alarms per patient-day to 0.77 at 2 per day, and the realised rates remain below the requested budgets. When the same A-calibrated threshold is applied to site B, the realised rate falls to roughly half the A rate and sensitivity at 1 per day falls from 0.585 to 0.315. The budget itself is still respected in the sense that B's realised mean is below the requested budget, but the clinical yield is much lower. This distinction matters for engineering practice: a site that imports only a threshold has not imported the operating point that the source site validated.

The comparison with the timestep shortcut clarifies what patient blocking buys, but not in the way a single-split table suggests. In ten repeated splits the shortcut stayed below the mean budget on both sites and often selected the same threshold as the plug-in rule, because its effective sample size is large. The data therefore do not show that timestep calibration fails empirically on this cohort. What they show is that the shortcut has no patient-level guarantee and that its patient-violation fraction is higher at 1 per day (0.296 versus 0.210). When the estimand is a patient-average rate, the effective sample size is the number of independent patients; a method that substitutes correlated hours can be less conservative without acquiring the guarantee it appears to have. The defensible conclusion is about the estimand and the audit trail, not a universal empirical ranking.

For engineering deployment, the results support five concrete practices. First, the minimum observed stay used in the support bound should be declared before looking at calibration or test outcomes, because it changes both eligibility and $L_{\max}$; the 24-hour restriction here removes about 22% of A records and 26% of B records. Second, the no-alarm fallback should remain an explicit, auditable state rather than being replaced by the largest finite threshold, since the latter can still alarm and has no certificate. Third, a target site that has local labels should recalibrate on its own patient blocks before adopting a source threshold, because the external rate and sensitivity can move in opposite directions; our oracle split shows that this recovers part of the sensitivity but not without a higher alarm load. A label-free alternative can match the requested rate on unlabelled target traces, a one-shot empirical quantile rule related to adaptive conformal inference [59], but it controls the alarm rate rather than the risk and therefore cannot replace a labelled certificate. Fourth, the threshold grid and the realised-rate report should be published together, since a grid floor can make high budgets look identical and hide the fact that the constraint is no longer binding. Fifth, a budget is a ceiling rather than a target: at loose budgets the selector should be allowed to stop before the budget is exhausted, because the marginal alarms are predominantly false and the official utility can fall below a tighter operating point. These practices are operational rather than architectural, and they are compatible with the alert-stewardship literature that treats interruptive alerts as a monitored resource [7, 8, 48].

The study has several limitations. It is retrospective and uses the two-site 2019 cohort plus the 2012 cohort as an outcome-shift stress test, so no independent same-task hospital cohort is available; the releases cannot link readmissions, so the independent-subject assumption remains an assumption. The U-CRC stage ranks calibration-feasible thresholds on a validation split, but it is a deployment heuristic and does not add a finite-sample utility guarantee. The label is the challenge's sepsis definition rather than an adjudicated deterioration event, left-censored records are excluded, and the score models are deliberately simple reference learners. The partial NEWS, SIRS and qSOFA baselines omit unavailable components, and the label-free adaptation rule is transductive and controls only the alarm rate. Target recalibration and label-budget results are oracle diagnostics that use target labels, not deployment procedures. No subgroup analysis or clinical outcome evaluation was performed, and ECE on a low-prevalence timestep problem is not evidence of clinical calibration. The coverage study uses ten splits per site, the GRU target diagnostic is single-seed, and no GPU-hour or alert-fatigue outcome was measured.

These limitations do not overturn the methodological result, but they change what should be claimed. The paper does not show that budgeted calibration improves patient outcomes, reduces mortality or is preferred by clinicians. It shows that a stateful policy can be calibrated against a patient-level alarm budget with a finite-sample bound, that the resulting operating curve is measurable on a public cohort, and that the same threshold does not transport unchanged across hospitals. That is a narrower contribution than a clinical-effectiveness claim, but it is a more honest one, and it identifies the target-calibration experiment that a deployment study would need to run next.

7. Conclusion

This paper treated early warning as a stateful alarm policy whose alert load is a declared design constraint. We derived a patient-blocked finite-grid upper-confidence procedure for the population mean of patient-level alarms per day, separated it from a deliberately invalid timestep shortcut, and implemented the path from causal scoring through policy replay to auditable operating points. On the 2019 A/B cohort, A sensitivity rose from 0.185 at 0.25 alarms per patient-day to 0.774 at 2 per day, but the same threshold did not transport: at 1 per day B sensitivity was 0.315 versus 0.585 at A, even though B's realised

rate stayed below the request. Target recalibration raised B sensitivity to 0.497 at a rate of 0.616, and 10% of eligible target subjects recovered the full-pool threshold at 1 per day. An independent 2012 mortality cohort confirmed that the budget remains portable while the transported operating point shifts, and the official 2019 utility showed that the best selector depends on whether the deployment values sensitivity or false-alarm cost. A validation-ranked CRC variant avoided over-alarming at the loose budget and raised official utility from 0.56 to 0.93 at 2 alarms per patient-day. These results support budgeted, patient-blocked calibration as an engineering instrument, not a clinical-effectiveness claim; prospective outcome evaluation and an independent same-task cohort remain necessary before deployment.

Data and code availability

The raw PhysioNet/CinC Challenge 2019 and 2012 records are publicly available from PhysioNet under their data use agreements and are not redistributed with this manuscript. The analysis code, subject-split audit, score-cache builders and experiment drivers are included in the accompanying repository. Aggregated result files, including the seed-level bootstrap outputs, the 2012 stress test, the official utility analysis and the alarm-queue study, are archived under `code/experiment_results/` and `results/`; the supplementary material is provided as `supplementary.pdf`. Raw records and score caches are not redistributed and can be regenerated from the public records with the documented commands.

Ethics statement

This study used only the public, de-identified PhysioNet/CinC Challenge 2019 cohort. No new patient data were collected, and no intervention or prospective clinical evaluation was performed. The analysis is a retrospective methodological study and does not constitute a clinical validation or a deployment recommendation.

CRediT authorship contribution statement

The author(s) contributed to the conceptualisation, methodology, software, validation, formal analysis, investigation, data curation, writing of the original draft, review and editing, and visualisation. All numerical results were produced by the documented code and checked against the archived result files.

Declaration of competing interest

The author(s) declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Ethical approval

This study used only the public, de-identified PhysioNet/CinC Challenge 2019 and 2012 cohorts. No new patient data were collected, and no intervention or prospective clinical evaluation was performed. The analysis is a retrospective methodological study and does not constitute a clinical validation or a deployment recommendation.

Use of generative artificial intelligence

Generative artificial-intelligence tools were used for language editing and code assistance. The author(s) verified all claims, references, numerical results and code, and take full responsibility for the content of the manuscript.

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